HIROFUMI INAGUMA

20230 Stevens Creek Blvd, Cupertino, CA, 95014, USA
Phone (US mobile): (+1) 650-705-8237 ⋄ (Japan mobile): (+81) 80-6967-1623
E-mail hiro.mhbc [at] gmail.com

Research scientist specializing in automatic speech recognition and speech translation, with contributions to large-scale multilingual systems (SeamlessM4T, Seamless) recognized globally, including TIME Best Inventions 2023 and publications in Nature and top-tier AI conferences.

RESEARCH INTERESTS

- Speech: Automatic speech recognition (ASR), speech translation, text-to-speech synthesis (TTS)
- Dialogue systems: Full-duplex speech LLM, social behavior
- Multimodality
- Codec Avatar: 3D human motion understanding/generation

EDUCATION

Ph.D. in Computer Science, Kyoto University, Kyoto, Japan *04/01/2018 - 08/31/2021*
Department of Intelligence Science and Technology, Graduate School of Informatics
- Thesis title: Fast and Low-Latency End-to-End Speech Recognition and Translation
- Supervisor: Prof. Tatsuya Kawahara
- Address: Yoshida-honmachi, Sakyo-ku, Kyoto, 606-8501, Japan

M.I. in Computer Science, Kyoto University, Kyoto, Japan *04/01/2016 - 03/31/2018*
Department of Intelligence Science and Technology, Graduate School of Informatics
- Thesis title: Joint Social Signal Detection and Automatic Speech Recognition based on End-to-End Modeling and Multi-task Learning
- Supervisor: Prof. Tatsuya Kawahara
- Address: Yoshida-honmachi, Sakyo-ku, Kyoto, 606-8501, Japan

B.E. in Computer Science, Kyoto University, Kyoto, Japan *04/01/2012 - 03/31/2016*
- Supervisor: Prof. Tatsuya Kawahara
- Address: Yoshida-honmachi, Sakyo-ku, Kyoto, 606-8501, Japan

WORK EXPERIENCES

Apple, Inc. *06/2026 - Current*
Senior Machine Learning Scientist at Apple
- Working on streaming ASR and full-duplex speech LLM.

Meta Platforms, Inc. *11/2023 - 03/2026*
Senior Research Scientist at FAIR and Reality Labs
- Developed systems enabling low-latency, real-time multimodal human-AI interaction at scale, improving responsiveness and user experience in conversational AI applications.
- Conducted large-scale training and optimization of full-duplex speech large language models (LLMs) for real-time conversational AI applications.
- Contributed to Seamless Interaction, a large-scale multimodal AI research initiative at Meta focused on real-time human-AI communication (`https://ai.meta.com/research/seamless-interaction/`).
- Designed and implemented low-latency 3D human motion generation models, enabling real-time synthesis of human body movements for interactive applications.
- Led the development of real-time audiovisual conversational systems, integrating speech, text, facial expressions, and 3D body motion representations for immersive human-AI interaction.
- Collaborated with cross-functional teams across FAIR and Reality Labs to advance state-of-the-art

research in speech, multimodal AI, and interactive systems.
- Address: 1 Hacker Way, Menlo Park, CA 94025, USA

Meta Platforms, Inc. *02/22/2022 - 11/05/2023*
Postdoctoral Research Scientist at Fundamental AI Research (FAIR)
- Conducted research and development on large-scale multilingual speech-to-speech translation systems, including SeamlessM4T and Seamless, enabling large-scale multilingual speech translation across approximately 100 languages in real-world applications (`https://ai.meta.com/research/seamless-communication`
- Designed and implemented UnitY, a novel end-to-end speech-to-speech translation architecture that served as a foundational backbone for SeamlessM4T.
- Designed and implemented UnitY2, an enhanced architecture that enabled multilingual speech translation models to operate in real-time streaming settings while preserving expressive speech output.
- Address: 1 Hacker Way, Menlo Park, CA 94025, USA

Microsoft Research, Redmond, WA, USA *07/2019 - 10/2019*
Research Internship (3 months)
- Mentor: Yifan Gong, Jinyu Li, Yashesh Gaur, Liang Lu
- Worked on low-emission-latency streaming end-to-end speech recognition

Johns Hopkins University, Baltimore, MD, USA *07/2018 - 09/2018*
Visiting student (2.5 months)
- Worked on end-to-end speech recognition and translation
- Participated in the JSALT workshop (topic: multilingual end-to-end speech recognition)
- Participated in the IWSLT2018 evaluation campaign on the end-to-end speech translation track
- Mentor: Prof. Shinji Watanabe

IBM Research AI, Tokyo, Japan *09/2017 - 11/2017*
Research Internship (2 months)
- Worked on end-to-end speech recognition
- Mentor: Gakuto Kurata, Takashi Fukuda

Recruit Co., Ltd., Tokyo, Japan *02/2017 - 02/2017*
Research Intern (2 weeks)
- Worked on data analysis and image classification competition
- Won the 1st place

CyberAgent, Inc., Tokyo, Japan *09/2016 - 10/2016*
Research Intern (1 month)
- Worked on the development of music recommendation systems

TECHNICAL STRENGTHS

Programming language	Python, Bash, C/C++, Java, LaTeX, PHP, JavaScript, CSS, HTML
Software & Tools	Fairseq, Fairseq2, ESPnet, Kaldi, Docker
Deep learning frameworks	PyTorch, TensorFlow

LANGUAGE SKILL

Japanese (native), English (fluent)

AWARDS

The Best Inventions of 2023, from TIME Magazine, October 2023
- Paper title: "*SeamlessM4T-Massively Multilingual & Multimodal Machine Translation*"
- `https://time.com/collections/best-inventions-2023/6326994/meta-seamlessm4t/`

Outstanding Paper Award, from Association for Computational Linguistics, July 2023
- Paper title: "*Hybrid Transducer and Attention based Encoder-Decoder Modeling for Speech-to-Text Tasks*"
- `https://2023.aclweb.org/program/best_papers/`

14th IEEE Signal Processing Society (SPS) Japan Student Conference Paper Award, from IEEE Signal Processing Society (SPS) Tokyo Joint Chapter, December 2020
- Paper title: "*Minimum Latency Training Strategies for Streaming Sequence-to-Sequence ASR*"
- `https://www.ieee-jp.org/section/tokyo/chapter/SP-01/past-student-paper.htm`

Microsoft Research Asia Ph.D. Fellowship Award (top 12 Ph.D. students in Asia), from Microsoft Research Asia (MSRA), October 2019
- `https://www.microsoft.com/en-us/research/academic-program/fellowships-microsoft-research-asia fellows/`

Yamashita SIG Research Award, from Information Processing Society of Japan (IPSJ), March 2019
- Paper title: "*An End-to-End Approach to Joint Social Signal Detection and Automatic Speech Recognition*"
- `https://www.ipsj.or.jp/award/yamasita2018-detail.html`

Yahoo! JAPAN award (best student paper), from SIG-SLP (Special Interest Group - Spoken Language Processing), June 2018

Full exemption from Repayment of Scholarship Loan for Students with Outstanding Results, from Japan Student Services Organization (JASSO), May 2018.

Research Fellowship for Young Scientists (DC1), from Japan Society for the Promotion of Science (JSPS), April 2018 - March 2021
- `https://kaken.nii.ac.jp/en/grant/KAKENHI-PROJECT-18J22864/`

Student award, from the Acoustical Society of Japan (ASJ), March 2018

Student award, from the 79th of National Convention of the Information Processing Society of Japan (IPSJ), March 2017

ACADEMIC SERVICES

Peer Review and Program Committee Service

Interspeech: 2021, 2022, 2023, 2025

ICASSP: 2019, 2020, 2021, 2023, 2026 (meta reviewer)

SLT: 2026 (meta reviewer)

ACL: 2023

EMNLP: 2023

ARR: 2024, 2025

IWSLT: 2021, 2023, 2024

CVPR: 2026

IEEE/ACM Transactions on Audio Speech and Language Processing

APSIPA Transactions on Signal and Information Processing

IEEE Signal Processing Letters

Program Committee

IWSLT: 2021, 2023

PUBLICATIONS (PREPRINT)

Vasu Agrawal et al., ”Seamless Interaction: Dyadic Audiovisual Motion Modeling and Large-Scale Dataset”, https://ai.meta.com/research/publications/seamless-interaction-dyadic-audiovisual-moti

Xutai Ma, Anna Sun, Siqi Ouyang, **Hirofumi Inaguma**, Paden Tomasello, ”Efficient Monotonic Multihead Attention”, https://arxiv.org/abs/2312.04515.

Seamless Communication, Loc Barrault, Yu-An Chung, Mariano Coria Meglioli, David Dale, Ning Dong, Mark Duppenthaler, Paul-Ambroise Duquenne, Brian Ellis, Hady Elsahar, Justin Haaheim, John Hoffman, Min-Jae Hwang, **Hirofumi Inaguma**, Christopher Klaiber, Ilia Kulikov, Pengwei Li, Daniel Licht, Jean Maillard, Ruslan Mavlyutov, Alice Rakotoarison, Kaushik Ram Sadagopan, Abinesh Ramakrishnan, Tuan Tran, Guillaume Wenzek, Yilin Yang, Ethan Ye, Ivan Evtimov, Pierre Fernandez, Cynthia Gao, Prangthip Hansanti, Elahe Kalbassi, Amanda Kallet, Artyom Kozhevnikov, Gabriel Mejia Gonzalez, Robin San Roman, Christophe Touret, Corinne Wong, Carleigh Wood, Bokai Yu, Pierre Andrews, Can Balioglu, Peng-Jen Chen, Marta R. Costa-juss, Maha Elbayad, Hongyu Gong, Francisco Guzmn, Kevin Heffernan, Somya Jain, Justine Kao, Ann Lee, Xutai Ma, Alex Mourachko, Benjamin Peloquin, Juan Pino, Sravya Popuri, Christophe Ropers, Safiyyah Saleem, Holger Schwenk, Anna Sun, Paden Tomasello, Changhan Wang, Jeff Wang, Skyler Wang, Mary Williamson, ”Seamless: Multilingual Expressive and Streaming Speech Translation”, https://arxiv.org/abs/2312.05187.

Seamless Communication, Loc Barrault, Yu-An Chung, Mariano Cora Meglioli, David Dale, Ning Dong, Paul-Ambroise Duquenne, Hady Elsahar, Hongyu Gong, Kevin Heffernan, John Hoffman, Christopher Klaiber, Pengwei Li, Daniel Licht, Jean Maillard, Alice Rakotoarison, Kaushik Ram Sadagopan, Guillaume Wenzek, Ethan Ye, Bapi Akula, Peng-Jen Chen, Naji El Hachem, Brian Ellis, Gabriel Mejia Gonzalez, Justin Haaheim, Prangthip Hansanti, Russ Howes, Bernie Huang, Min-Jae Hwang, **Hirofumi Inaguma**, Somya Jain, Elahe Kalbassi, Amanda Kallet, Ilia Kulikov, Janice Lam, Daniel Li, Xutai Ma, Ruslan Mavlyutov, Benjamin Peloquin, Mohamed Ramadan, Abinesh Ramakrishnan, Anna Sun, Kevin Tran, Tuan Tran, Igor Tufanov, Vish Vogeti, Carleigh Wood, Yilin Yang, Bokai Yu, Pierre Andrews, Can Balioglu, Marta R. Costa-juss, Onur Celebi, Maha Elbayad, Cynthia Gao, Francisco Guzmn, Justine Kao, Ann Lee, Alexandre Mourachko, Juan Pino, Sravya Popuri, Christophe Ropers, Safiyyah Saleem, Holger Schwenk, Paden Tomasello, Changhan Wang, Jeff Wang, Skyler Wang, ”SeamlessM4T-Massively Multilingual & Multimodal Machine Translation”, https://arxiv.org/abs/2308.11596.

Hayato Futami, **Hirofumi Inaguma**, Masato Mimura, Shinsuke Sakai, Tatsuya Kawahara, ”Distilling the Knowledge of BERT for CTC-based ASR”, https://arxiv.org/abs/2209.02030.

Hirofumi Inaguma, Yosuke Higuchi, Kevin Duh, Tatsuya Kawahara, Shinji Watanabe, ”Non-autoregressive End-to-end Speech Translation with Parallel Autoregressive Rescoring”, *IEEE/ACM Transactions on Audio, Speech, Language Processing*, https://arxiv.org/abs/2109.04411.

PUBLICATIONS (PEER-REVIEWED JOURNAL PAPER)

[Nature] Seamless Communication Team, ”Joint speech and text machine translation for up to 100 languages”, https://pmc.ncbi.nlm.nih.gov/articles/PMC11735396/.

[TASLP] **Hirofumi Inaguma**, Tatsuya Kawahara, ”Alignment Knowledge Distillation for Online Streaming Attention-based Speech Recognition”, IEEE/ACM Transactions on Audio, Speech, Language Processing, https://arxiv.org/abs/2103.00422.

PUBLICATIONS (PEER-REVIEWED INTERNATIONAL CONFERENCE PAPER)

[EMNLP2025 (findings)] Weiting Tan, Jiachen Lian, **Hirofumi Inaguma**, Paden Tomasello, Philipp Koehn, Xutai Ma, "Seeing is Believing: Emotion-Aware Audio-Visual Language Modeling for Expressive Speech Generation", EMNLP, findings, 2025, `https://arxiv.org/abs/2508.16188`.

[IWSLT2025] Weiting Tan, **Hirofumi Inaguma**, Ning Dong, Paden Tomasello, Xutai Ma, "SSR: Alignment-Aware Modality Connector for Speech Language Models", IWSLT, 2025, `https://arxiv.org/abs/2410.00168`.

[SLT2024] **Hirofumi Inaguma**, Ilia Kulikov, Zhaoheng Ni, Sravya Popuri, Paden Tomasello, "Massively Multilingual Forced Aligner Leveraging Self-supervised Discrete Units", IEEE SLT, pp. 899-905, 2024, `https://ieeexplore.ieee.org/abstract/document/10832291`.

[Interspeech2024] Chao-Wei Huang, Hui Lu, Hongyu Gong, **Hirofumi Inaguma**, Ilia Kulikov, Ruslan Mavlyutov, Sravya Popuri, "Investigating Decoder-only Large Language Models for Speech-to-text Translation", Interspeech, pp. 832-836, 2024, `https://arxiv.org/abs/2407.03169`.

[Interspeech2024] Jiatong Shi, Xutai Ma, **Hirofumi Inaguma**, Anna Sun, Shinji Watanabe, "MMM: Multi-Layer Multi-Residual Multi-Stream Discrete Speech Representation from Self-supervised Learning Model", Interspeech, pp. 2569-2573, 2024, `https://arxiv.org/abs/2406.09869`.

[ICLR2024] Jiatong Shi, **Hirofumi Inaguma**, Xutai Ma, Ilia Kulikov, Anna Sun, "Multi-resolution HuBERT: Multi-resolution Speech Self-Supervised Learning with Masked Unit Prediction", ICLR, 2024, `https://arxiv.org/abs/2310.02720`. (**Spotlight**)

[Interspeech2023] Jiatong Shi, Yun Tang, **Hirofumi Inaguma**, Hongyu Gong, Juan Pino, Shinji Watanabe, "Exploration on HuBERT with Multiple Resolutions", Interspeech, pp. 3287-3291, 2023, `https://arxiv.org/abs/2306.01084`.

[IWSLT2023] Milind Agarwal, Sweta Agrawal, Antonios Anastasopoulos, Luisa Bentivogli, Ondej Bojar, Claudia Borg, Marine Carpuat, Roldano Cattoni, Mauro Cettolo, Mingda Chen, William Chen, Khalid Choukri, Alexandra Chronopoulou, Anna Currey, Thierry Declerck, Qianqian Dong, Kevin Duh, Yannick Estve, Marcello Federico, Souhir Gahbiche, Barry Haddow, Benjamin Hsu, Phu Mon Htut, **Hirofumi Inaguma**, Dvid Javorsk, John Judge, Yasumasa Kano, Tom Ko, Rishu Kumar, Pengwei Li, Xutai Ma, Prashant Mathur, Evgeny Matusov, Paul McNamee, John P. McCrae, Kenton Murray, Maria Nadejde, Satoshi Nakamura, Matteo Negri, Ha Nguyen, Jan Niehues, Xing Niu, Atul Kr. Ojha, John E. Ortega, Proyag Pal, Juan Pino, Lonneke van der Plas, Peter Polk, Elijah Rippeth, Elizabeth Salesky, Jiatong Shi, Matthias Sperber, Sebastian Stker, Katsuhito Sudoh, Yun Tang, Brian Thompson, Kevin Tran, Marco Turchi, Alex Waibel, Mingxuan Wang, Shinji Watanabe, Rodolfo Zevallos, "Findings of the IWSLT 2023 evaluation campaign", IWSLT, pp. 1-61, 2023, `https://aclanthology.org/2023.iwslt-1.1/`.

[ACL2023 (main)] **Hirofumi Inaguma**, Sravya Popuri, Ilia Kulikov, Peng-Jen Chen, Changhan Wang, Yu-An Chung, Yun Tang, Ann Lee, Shinji Watanabe, Juan Pino, "UnitY: Two-pass Direct Speech-to-speech Translation with Discrete Units", ACL, main conference, pp. 15655-15680, 2023, `https://arxiv.org/abs/2212.08055`.

[ACL2023 (main)] Yun Tang, Anna Y. Sun, **Hirofumi Inaguma**, Xinyue Chen, Ning Dong, Xutai Ma, Paden D. Tomasello, Juan Pino, "Hybrid Transducer and Attention based Encoder-Decoder Modeling for Speech-to-Text Tasks", ACL, main conference, pp. 12441-12455, 2023, `https://arxiv.org/abs/2305.03101`. (**Outstanding Paper Award**)

[ACL2023 (main)] Changhan Wang, **Hirofumi Inaguma**, Peng-Jen Chen, Ilia Kulikov, Yun Tang, Wei-Ning Hsu, Michael Auli, Juan Pino, "Simple and Effective Unsupervised Speech Translation", ACL, main conference, pp. 10771-10784, 2023, `https://arxiv.org/abs/2210.10191`.

[ACL2023 (findings)] Peng-Jen Chen, Kevin Tran, Yilin Yang, Jingfei Du, Justine Kao, Yu-An Chung, Paden Tomasello, Paul-Ambroise Duquenne, Holger Schwenk, Hongyu Gong, **Hirofumi Inaguma**, Sravya Popuri, Changhan Wang, Juan Pino, Wei-Ning Hsu, Ann Lee, "Speech-to-Speech Translation For A Real-world Unwritten Language", ACL, findings, pp. 4969-4983, 2023, https://arxiv.org/abs/2211.06474.

[ACL2023 (demo)] Brian Yan, Jiatong Shi, Yun Tang, **Hirofumi Inaguma**, Yifan Peng, Siddharth Dalmia, Peter Polk, Patrick Fernandes, Dan Berrebbi, Tomoki Hayashi, Xiaohui Zhang, Zhaoheng Ni, Moto Hira, Soumi Maiti, Juan Pino, Shinji Watanabe, "ESPnet-ST-v2: Multipurpose Spoken Language Translation Toolkit", ACL, System Demonstrations, pp. 400-411, 2023, https://arxiv.org/abs/2304.04596.

[ICASSP2023] Marco Gaido, Yun Tang, Ilia Kulikov, Rongqing Huang, Hongyu Gong, **Hirofumi Inaguma**, "Named Entity Detection and Injection for Direct Speech Translation", IEEE ICASSP, pp. 1-5, 2023, https://arxiv.org/abs/2210.11981.

[ICASSP2023] Jiatong Shi, Yun Tang, Ann Lee, **Hirofumi Inaguma**, Changhan Wang, Juan Pino, Shinji Watanabe, "Enhancing Speech-to-Speech Translation with Multiple TTS Targets", IEEE ICASSP, pp. 1-5, 2023, https://arxiv.org/abs/2304.04618.

[Interspeech2022] Hayato Futami, **Hirofumi Inaguma**, Sei Ueno, Masato Mimura, Shinsuke Sakai, Tatsuya Kawahara, "Non-autoregressive Error Correction for CTC-based ASR with Phone-conditioned Masked LM", Interspeech, pp. 3889-3893, 2022, https://arxiv.org/abs/2209.04062.

[ASRU2021] **Hirofumi Inaguma**, Siddharth Dalmia, Brian Yan, Shinji Watanabe, "Fast-MD: Fast Multi-Decoder End-to-End Speech Translation with Non-Autoregressive Hidden Intermediates", IEEE ASRU, pp. 922-929, 2021, https://arxiv.org/abs/2109.12804. (acceptance rate: 157/330=47.5%).

[ASRU2021] Yosuke Higuchi, Nanxin Chen, Yuya Fujita, **Hirofumi Inaguma**, Tatsuya Komatsu, Jaesong Lee, Jumon Nozaki, Tianzi Wang, Shinji Watanabe, "A COMPARATIVE STUDY ON NON-AUTOREGRESSIVE MODELINGS FOR SPEECH-TO-TEXT GENERATION", IEEE ASRU, pp. 47-54, 2021, https://arxiv.org/abs/2110.05249.

[ASRU2021] Florian Boyer, Yusuke Shinohara, Takaaki Ishii, **Hirofumi Inaguma**, Shinji Watanabe, "A STUDY OF TRANSDUCER BASED END-TO-END ASR WITH ESPNET: ARCHITECTURE, AUXILIARY LOSS AND DECODING STRATEGIES", IEEE ASRU, pp. 16-23, 2021, https://arxiv.org/abs/2201.05420.

[ASRU2021] Hayato Futami, **Hirofumi Inaguma**, Masato Mimura, Shinsuke Sakai, Tatsuya Kawahara, "ASR RESCORING AND CONFIDENCE ESTIMATION WITH ELECTRA", IEEE ASRU, pp. 380-387, 2021, https://arxiv.org/abs/2110.01857.

[IWSLT2021 (system)] **Hirofumi Inaguma***, Brian Yan*, Siddharth Dalmia, Pengcheng Guo, Jiatong Shi, Kevin Duh, Shinji Watanabe (*equal contribution), "ESPnet-ST IWSLT 2021 Offline Speech Translation System", IWSLT, 2021, https://arxiv.org/abs/2107.00636.

[Interspeech2021] **Hirofumi Inaguma** and Tatsuya Kawahara, "StableEmit: Selection Probability Discount for Reducing Emission Latency of Streaming Monotonic Attention ASR", Interspeech, pp. 1817-1821, 2021, https://arxiv.org/abs/2107.00635. (acceptance rate: 963/1990=48.4%)

[Interspeech2021] **Hirofumi Inaguma** and Tatsuya Kawahara, "VAD-free Streaming Hybrid CTC/Attention ASR for Unsegmented Recording", Interspeech, pp. 4049-4053, 2021, https://arxiv.org/abs/2107.07509.

[NAACL-HLT2021] **Hirofumi Inaguma**, Tatsuya Kawahara, Shinji Watanabe, "Source and Target Bidirectional Knowledge Distillation for End-to-end Speech Translation", NAACL-HLT, pp. 1872-1881, 2021, https://arxiv.org/abs/2104.06457. (acceptance rate: 23%)

[DSLW2021] Shinji Watanabe, Florian Boyer, Xuankai Chang, Pengcheng Guo, Tomoki Hayashi, Yosuke Higuchi, Takaaki Hori, Wen-Chin Huang, **Hirofumi Inaguma**, Naoyuki Kamo, Shigeki Karita, Chenda Li, Jing Shi, Aswin Shanmugam Subramanian, Wangyou Zhang, "The 2020 ESPnet update: new features, broadened applications, performance improvements, and future plans", IEEE Data Science and Learning Workshop (DSLW), 2021, https://arxiv.org/abs/2012.13006.

[ICASSP2021] **Hirofumi Inaguma**, Yosuke Higuchi, Kevin Duh, Tatsuya Kawahara, Shinji Watanabe, "Orthros: Non-autoregressive End-to-end Speech Translation with Dual-decoder", IEEE ICASSP, pp. 7488-7492, 2021, https://arxiv.org/abs/2010.13047. (acceptance rate: 1734/3610=48.0%)

[ICASSP2021] Yosuke Higuchi, **Hirofumi Inaguma**, Shinji Watanabe, Tetsuji Ogawa, Tetsunori Kobayashi, "Improved Mask-CTC for Non-Autoregressive End-to-End ASR", IEEE ICASSP, pp. 3655-3659, 2021, https://arxiv.org/abs/2010.13270.

[ICASSP2021] Pengcheng Guo, Florian Boyer, Xuankai Chang, Tomoki Hayashi, Yosuke Higuchi, **Hirofumi Inaguma**, Naoyuki Kamo, Chenda Li, Daniel Garcia-Romero, Jiatong Shi, Jing Shi, Shinji Watanabe, Kun Wei, Wangyou Zhang, Yuekai Zhang, "Recent Developments on ESPnet Toolkit Boosted by Conformer", IEEE ICASSP, pp. 5874-5878, 2021, https://arxiv.org/abs/2010.13956.

[Interspeech2020] **Hirofumi Inaguma**, Masato Mimura, Tatsuya Kawahara, "CTC-synchronous Training for Monotonic Attention Model", Interspeech, pp. 571-575, 2020, https://arxiv.org/abs/2005.04712. (acceptance rate: 47.0%)

[Interspeech2020] **Hirofumi Inaguma**, Masato Mimura, Tatsuya Kawahara, "Enhancing Monotonic Multihead Attention for Streaming ASR", Interspeech, pp. 2137-2141, 2020, https://arxiv.org/abs/2005.09394.

[Interspeech2020] Hayato Futami, **Hirofumi Inaguma**, Sei Ueno, Masato Mimura, Shinsuke Sakai, Tatsuya Kawahara, "Distilling the Knowledge of BERT for Sequence-to-Sequence ASR", Interspeech, pp. 3635-3639, 2020, https://arxiv.org/abs/2008.03822.

[Interspeech2020] Trung V. Dang, Tianyu Zhao, Sei Ueno, **Hirofumi Inaguma**, Tatsuya Kawahara, "End-to-end speech-to-dialog-act recognition", Interspeech, pp. 3910-3914, 2020, https://arxiv.org/abs/2004.11419.

[ACL2020 (demo)] **Hirofumi Inaguma**, Shun Kiyono, Kevin Duh, Shigeki Karita, Nelson Yalta, Tomoki Hayashi, Shinji Watanabe, "ESPnet-ST: All-in-One Speech Translation Toolkit", ACL, System Demonstrations, pp. 302-311, 2020, https://arxiv.org/abs/2004.10234.

[ICASSP2020] **Hirofumi Inaguma**, Yashesh Gaur, Liang Lu, Jinyu Li, Yifan Gong, "Minimum latency training strategies for streaming sequence-to-sequence ASR", IEEE ICASSP, pp. 6064-6068, 2020, https://arxiv.org/abs/2004.05009. (acceptance rate: 47%, **Oral**)

[ASRU2019] **Hirofumi Inaguma**, Kevin Duh, Tatsuya Kawahara, Shinji Watanabe, "Multilingual End-To-End Speech Translation", IEEE ASRU, pp. 570-577, 2019, https://arxiv.org/abs/1910.00254. (acceptance rate: 144/299=48.1%)

[ASRU2019] Shigeki Karita, Nanxin Chen, Tomoki Hayashi, Takaaki Hori, **Hirofumi Inaguma**, Ziyan Jiang, Masao Someki, Nelson Enrique Yalta Soplin, Ryuichi Yamamoto, Xiaofei Wang, Shinji Watanabe, Takenori Yoshimura, Wangyou Zhang, "A Comparative Study on Transformer vs RNN in Speech Applications", IEEE ASRU, pp. 449-456, 2019, https://arxiv.org/abs/1909.06317.

[ICASSP2019] **Hirofumi Inaguma**, Jaejin Cho, Murali Karthick Baskar, Tatsuya Kawahara, Shinji Watanabe, "Transfer Learning of Language-Independent End-to-End ASR with Language Model Fusion", IEEE ICASSP, pp. 6096-6100, 2019, https://arxiv.org/abs/1811.02134. (acceptance rate: 1774/3815=46.5%)

[ICASSP2019] Jaejin Cho, Shinji Watanabe, Takaaki Hori, Murali Karthick Baskar, **Hirofumi Inaguma**, Jesus Villalba, Najim Dehak, "Language Model Integration Based on Memory Control for Sequence to

Sequence Speech Recognition", IEEE ICASSP, pp. 6191-6195, 2019, https://arxiv.org/abs/1811.02162.

[SLT2018] **Hirofumi Inaguma**, Masato Mimura, Shinsuke Sakai, Tatsuya Kawahara, "Improving OOV Detection and Resolution with External Language Models in Acoustic-to-Word ASR", IEEE SLT, pp. 212-218, 2018, http://sap.ist.i.kyoto-u.ac.jp/EN/bib/intl/INA-SLT18.pdf. (acceptance rate: 150/257=58.3%)

[SLT2018] Masato Mimura, Sei Ueno, **Hirofumi Inaguma**, Shinsuke Sakai, Tatsuya Kawahara, "Leveraging Sequence-to-Sequence Speech Synthesis for Enhancing Acoustic-to-Word Speech Recognition", IEEE SLT, pp. 477-484, 2018, http://sap.ist.i.kyoto-u.ac.jp/lab/bib/intl/MIM-SLT18.pdf.

[IWSLT2018 (system)] **Hirofumi Inaguma**, Xuan Zhang, Zhiqi Wang, Adithya Renduchintala, Shinji Watanabe, Kevin Duh, "The JHU/KyotoU Speech Translation System for IWSLT 2018", IWSLT, 2018.

[ICASSP2018] **Hirofumi Inaguma**, Masato Mimura, Koji Inoue, Kazuyoshi Yoshii, Tatsuya Kawahara, "An End-to-End Approach to Joint Social Signal Detection and Automatic Speech Recognition", IEEE ICASSP, pp. 6214-6218, 2018, http://sap.ist.i.kyoto-u.ac.jp/EN/bib/intl/INA-ICASSP18.pdf. (acceptance rate: 1406/2830=49.7%)

[ICASSP2018] Sei Ueno, **Hirofumi Inaguma**, Masato Mimura, Tatsuya Kawahara, "Acoustic-to-Word Attention-Based Model Complemented with Character-level CTC-Based Model", IEEE ICASSP, pp. 5804-5808, 2018, http://www.sap.ist.i.kyoto-u.ac.jp/lab/bib/intl/UEN-ICASSP18.pdf.

[Interspeech2017] **Hirofumi Inaguma**, Koji Inoue, Masato Mimura, Tatsuya Kawahara, "Social Signal Detection in Spontaneous Dialogue Using Bidirectional LSTM-CTC", Interspeech, pp. 1691-1695, 2017. (acceptance rate: 799/1582=52.0%)

[ICMI2016 WS] **Hirofumi Inaguma**, Koji Inoue, Shizuka Nakamura, Katsuya Takanashi, Tatsuya Kawahara, "Prediction of Ice-breaking between participants using prosodic features in the first meeting dialogue", International Conference Multimodal Interaction workshop on Advancements in Social Signal Processing for Multimodal Interaction (ASSP4MI), 2016.

(Last update: 9/10/2026)